

Container Orchestration Tool – Kubernetes

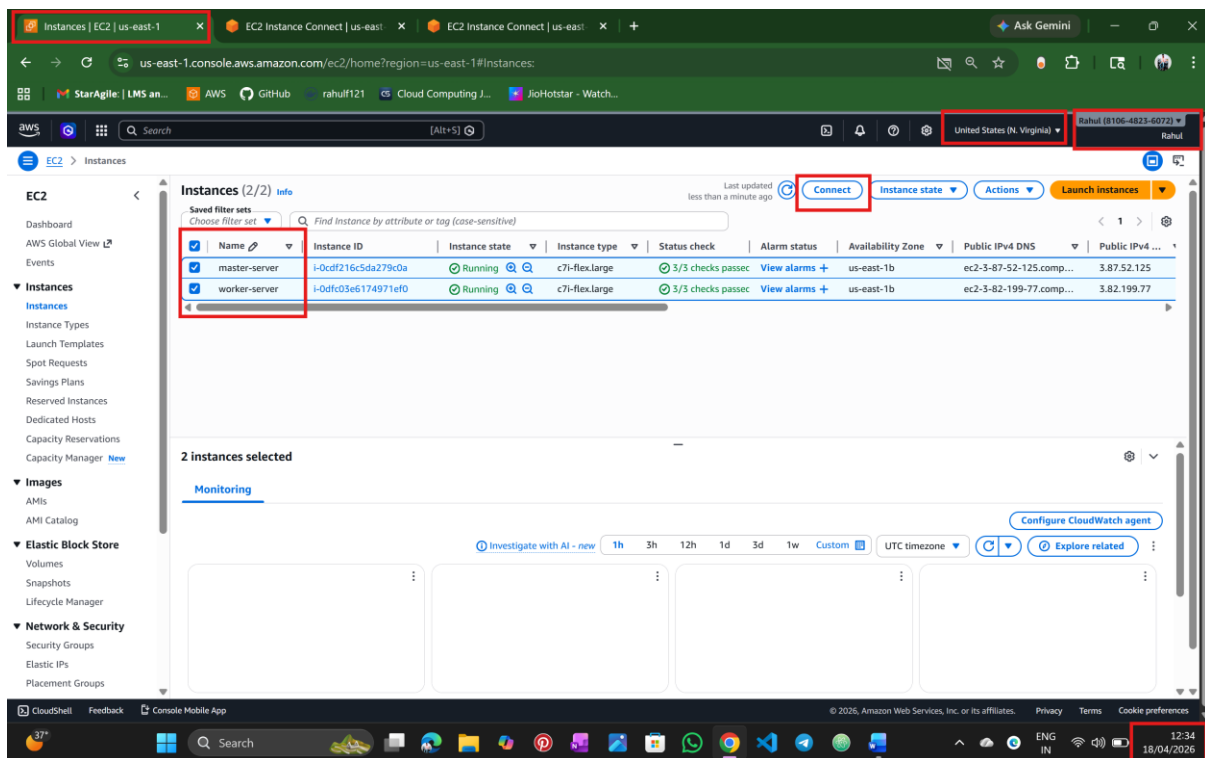
Submitted to - Vikul Sir

Date of Submission – 19/APR/2026

Submitted by – Rahul Nalathathi

L1 - Create Deployment Controller Object to Deploy the Application Image Created in Docker Module and Expose it to the Internet

Step :1 Login into AWS and create two ec2 instances



Step :2 Connect master node and worker node with token and install prerequisites

The screenshot shows the AWS Management Console for an EC2 Instance Connect session. The browser tabs include 'Instances | EC2 | us-east-1', 'EC2 Instance Connect | us-east-1', and 'EC2 Instance Connect | us-east-1'. The address bar shows the URL: `us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=us-east-1&connType=standard&instanceId=i-0dfc03e6174971ef0&osUser=...`. The AWS region is set to 'United States (N. Virginia)'. The user profile is 'Rahul (8106-432-4072)'. The terminal window shows the command `kubectl get nodes` and its output:

NAME	STATUS	ROLES	AGE	VERSION
ip-172-31-90-206.ec2.internal	Ready	control-plane	3m10s	v1.29.15
ip-172-31-93-203.ec2.internal	Ready	<none>	30s	v1.29.15

The instance ID is `i-0dfc03e6174971ef0 (worker-server)`. The public IP is `3.82.199.77` and the private IP is `172.31.90.206`. The bottom status bar shows the time as 12:46 on 18/04/2026.

The screenshot shows the AWS Management Console for an EC2 Instance Connect session. The browser tabs include 'Instances | EC2 | us-east-1', 'EC2 Instance Connect | us-east-1', and 'EC2 Instance Connect | us-east-1'. The address bar shows the URL: `us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?addressFamily=ipv4&connType=standard&instanceId=i-0cdf216c5da279c0a&osUser=...`. The AWS region is set to 'United States (N. Virginia)'. The user profile is 'Rahul (8106-432-4072)'. The terminal window shows the command `kubeadm join` and its output:

```
Verifying : kubectl-1.29.15-150500.1.1.x86_64
Verifying : kubelet-1.29.15-150500.1.1.x86_64
Verifying : kubernetes-cni-1.3.0-150500.1.1.x86_64

Installed:
conntrack-tools-1.4.6-2.amzn2023.0.2.x86_64      cri-tools-1.29.0-150500.1.1.x86_64      iptables-libs-1.0.8-3.amzn2023.0.2.x86_64
iptables-nft-1.0.8-3.amzn2023.0.2.x86_64      kubeadm-1.29.15-150500.1.1.x86_64      kubectl-1.29.15-150500.1.1.x86_64
kubernetes-cni-1.3.0-150500.1.1.x86_64      kubenetes-cni-1.3.0-150500.1.1.x86_64      libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64
libnetfilter_cthelper-1.0.0-21.amzn2023.0.2.x86_64  libnetfilter_cttimeout-1.0.0-19.amzn2023.0.2.x86_64  libnetfilter_guose-1.0.5-2.amzn2023.0.2.x86_64
libnftnl-1.0.1-19.amzn2023.0.2.x86_64      libnftnl-1.2.2-2.amzn2023.0.2.x86_64

Complete!
[ec2-user@ip-172-31-93-203 ~]$ sudo systemctl enable --now kubelet
Created symlink /etc/systemd/system/multi-user.target.wants/kubelet.service to /usr/lib/systemd/system/kubelet.service.
[ec2-user@ip-172-31-93-203 ~]$ kubeadm join 172.31.90.206:6443 --token pih3r37w9t1a15qysc8aoo --discovery-token-ca-cert-hash sha256:42ba238fc914a42764cdec1763ccf464d148497d2da44c18d26bf7fd619
86d7
[ec2-user@ip-172-31-93-203 ~]$
error execution phase preflight: [preflight] Some fatal errors occurred:
[ERROR IsPrivilegedUser]: user is not running as root
[preflight] If you know what you are doing, you can make a check non-fatal with '--ignore-preflight-errors=...'
To see the stack trace of this error execute with '--v=5 or higher
[ec2-user@ip-172-31-93-203 ~]$ sudo kubeadm join 172.31.90.206:6443 --token pih3r37w9t1a15qysc8aoo \
--discovery-token-ca-cert-hash sha256:42ba238fc914a42764cdec1763ccf464d148497d2da44c18d26bf7fd619d6d7
[preflight] Running pre-flight checks
[WARNING FileExisting-tc]: tc not found in system path
[preflight] Reading configuration from the cluster...
[preflight] FYI: You can look at this config file with 'kubectl -n kube-system get cm kubeadm-config -o yaml'
[kubelet-start] Writing kubelet configuration to file "/var/lib/kubelet/config.yaml"
[kubelet-start] Writing kubelet environment file with flags to file "/var/lib/kubelet/kubeadm-flags.env"
[kubelet-start] Starting the kubelet
[kubelet-start] Waiting for the kubelet to perform the TLS Bootstrap...

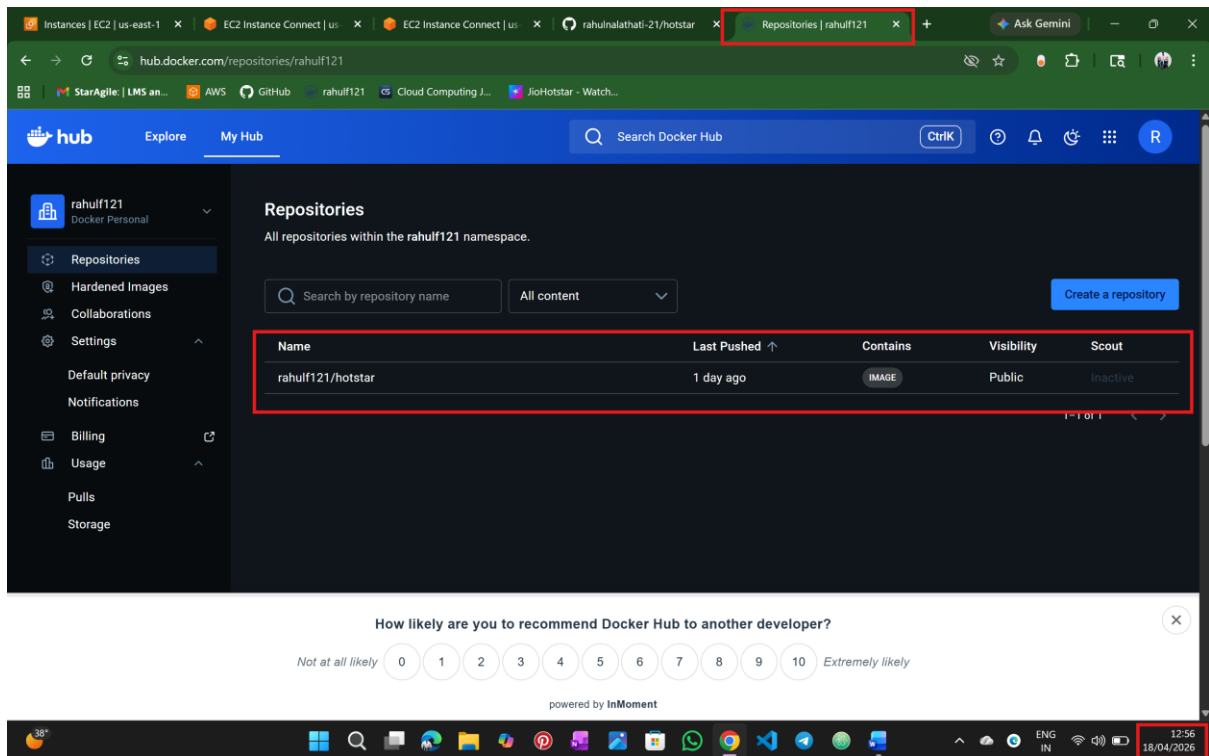
This node has joined the cluster:
* Certificate signing request was sent to apiserer and a response was received.
* The Kubelet was informed of the new secure connection details.

Run 'kubectl get nodes' on the control-plane to see this node join the cluster.
[ec2-user@ip-172-31-93-203 ~]$
```

The instance ID is `i-0cdf216c5da279c0a (master-server)`. The public IP is `3.87.52.125` and the private IP is `172.31.93.203`. The bottom status bar shows the time as 12:47 on 18/04/2026.

Step :3 I already have docker image in my docker hub and used.

→if not push docker image to docker hub



Step :4 Now create a folder and go to that folder and create

→deployment.yaml

→service.yaml

Instances | EC2 | us-east-1 x EC2 Instance Connect | us x EC2 Instance Connect | us x rahuinalathati-21/hotstar x Repositories | rahulf121 x Ask Gemini

us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=us-east-1&connType=standard&instanceId=i-0dfc03e6174971ef0&osUser=...

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aws Search [Alt+S] Ask Amazon Q United States (N. Virginia) Rahul (8106-4923-4072) Rahul

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: hotstar-deployment
spec:
  replicas: 2
  selector:
    matchLabels:
      app: hotstar
  template:
    metadata:
      labels:
        app: hotstar
    spec:
      containers:
        - name: hotstar-container
          image: rahulf121/hotstar:latest
          ports:
            - containerPort: 8080
```

15,17 A11

i-0dfc03e6174971ef0 (worker-server)
PublicIPs: 3.82.199.77 PrivateIPs: 172.31.90.206

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38° ENG IN 12:57 18/04/2026

Instances | EC2 | us-east-1 x EC2 Instance Connect | us x EC2 Instance Connect | us x rahuinalathati-21/hotstar x Repositories | rahulf121 x Ask Gemini

us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=us-east-1&connType=standard&instanceId=i-0dfc03e6174971ef0&osUser=...

StarAgile | LMS an... AWS GitHub rahulf121 Cloud Computing J... JioHotstar - Watch...

aws Search [Alt+S] Ask Amazon Q United States (N. Virginia) Rahul (8106-4923-4072) Rahul

```
apiVersion: v1
kind: Service
metadata:
  name: hotstar-service
spec:
  type: NodePort
  selector:
    app: hotstar
  ports:
    - protocol: TCP
      port: 80
      targetPort: 8080
      nodePort: 30000
```

13,21 A11

i-0dfc03e6174971ef0 (worker-server)
PublicIPs: 3.82.199.77 PrivateIPs: 172.31.90.206

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38° ENG IN 12:59 18/04/2026

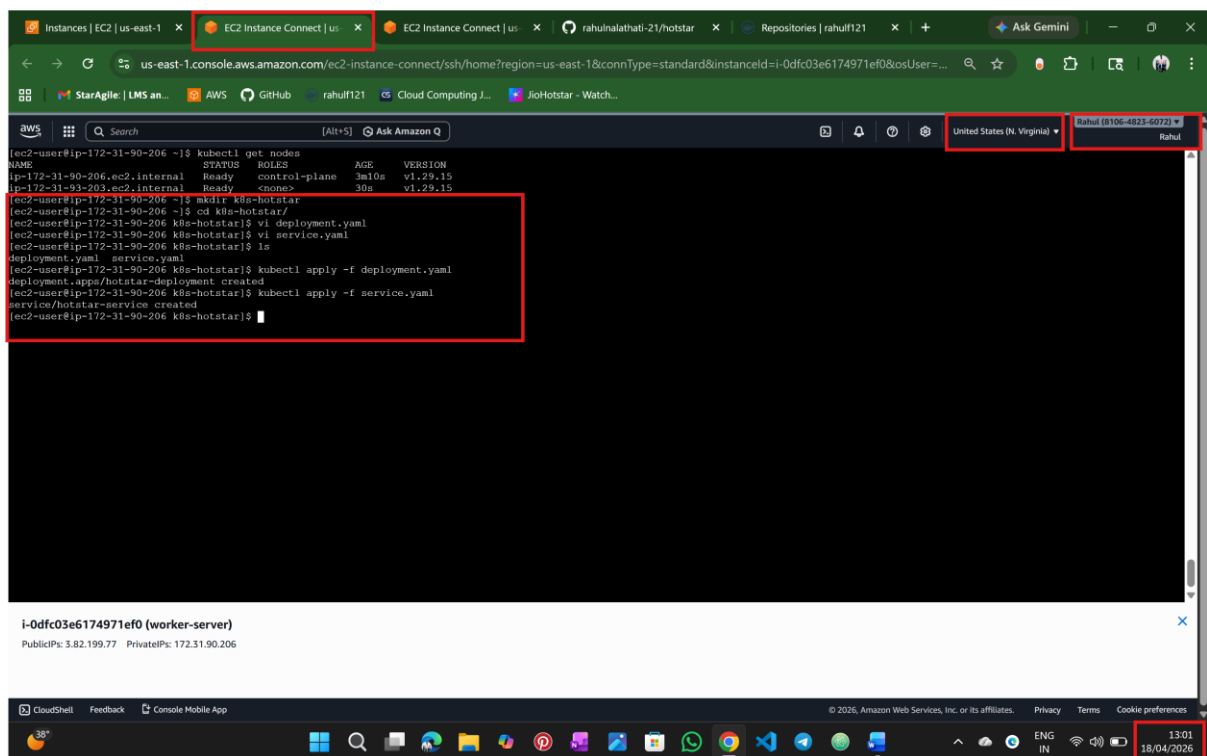
Step :5 Apply and Verify

→ kubectl apply -f deployment.yaml

→ kubectl apply -f service.yaml

→ kubectl get pods

→ kubectl get svc

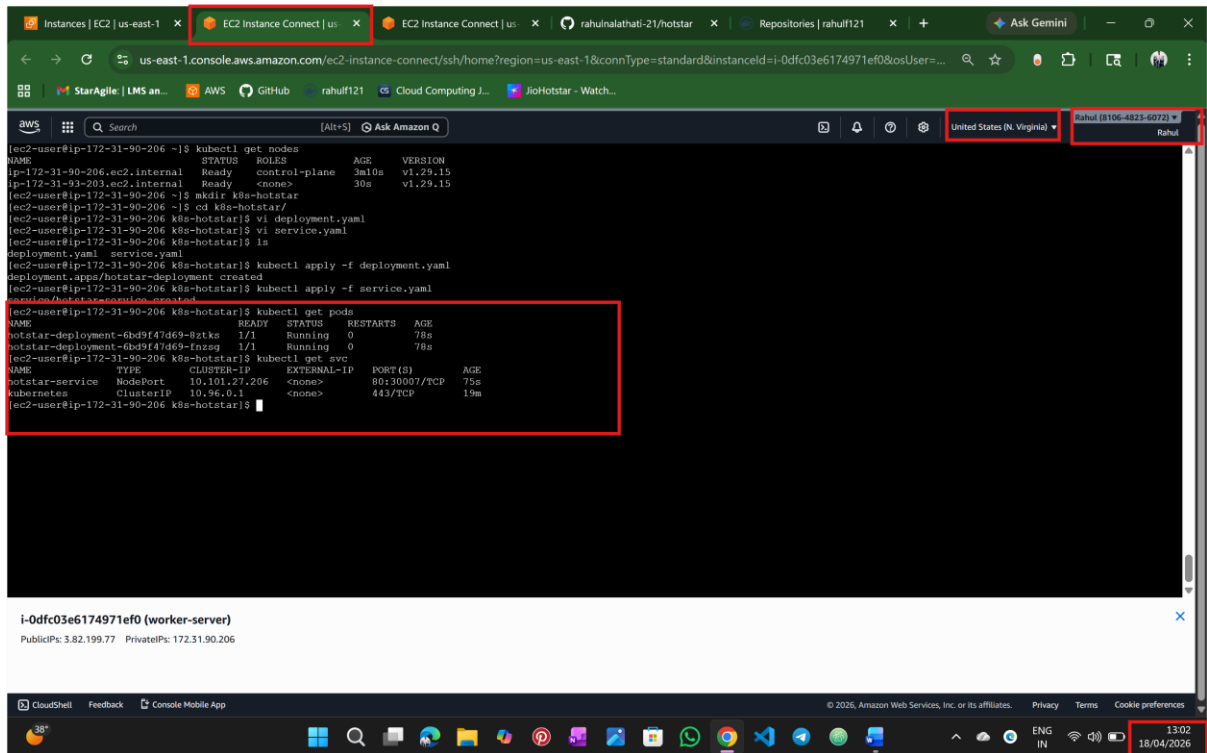


The screenshot shows a web browser window with multiple tabs, including 'Instances | EC2 | us-east-1', 'EC2 Instance Connect | us...', and 'rahulalathati-21/hotstar'. The active tab displays a terminal session for an EC2 instance named 'i-0dfc03e6174971ef0 (worker-server)'. The terminal output shows the following commands and results:

```
[ec2-user@ip-172-31-90-206 ~]$ kubectl get nodes
NAME                                STATUS    ROLES    AGE   VERSION
ip-172-31-90-206.ec2.internal      Ready    control-plane   3m10s   v1.29.15
ip-172-31-93-203.ec2.internal      Ready    <none>         30s     v1.29.15

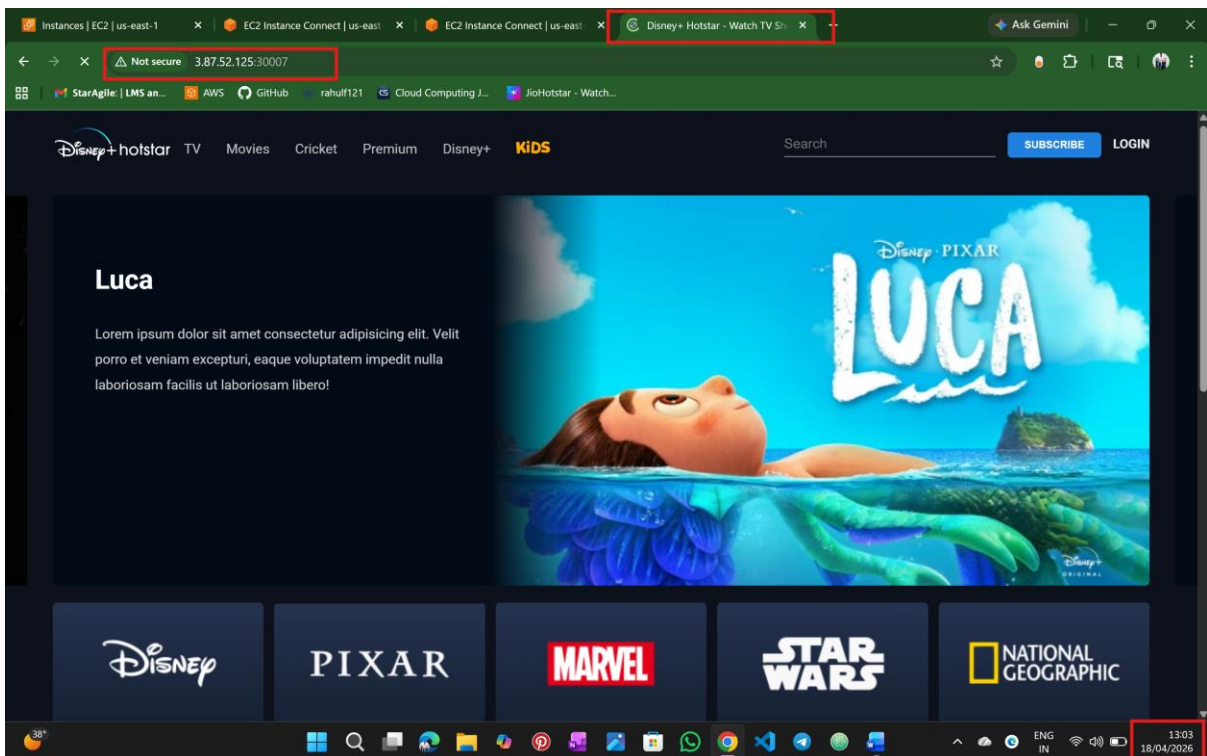
[ec2-user@ip-172-31-90-206 ~]$ mkdir k8s-hotstar
[ec2-user@ip-172-31-90-206 ~]$ cd k8s-hotstar/
[ec2-user@ip-172-31-90-206 k8s-hotstar]$ vi deployment.yaml
[ec2-user@ip-172-31-90-206 k8s-hotstar]$ vi service.yaml
[ec2-user@ip-172-31-90-206 k8s-hotstar]$ ls
deployment.yaml  service.yaml
[ec2-user@ip-172-31-90-206 k8s-hotstar]$ kubectl apply -f deployment.yaml
deployment.apps/hotstar-deployment created
[ec2-user@ip-172-31-90-206 k8s-hotstar]$ kubectl apply -f service.yaml
service/hotstar-service created
[ec2-user@ip-172-31-90-206 k8s-hotstar]$
```

Below the terminal output, the instance details are shown: 'i-0dfc03e6174971ef0 (worker-server)' with PublicIPs: 3.82.199.77 and PrivateIPs: 172.31.90.206. The bottom of the screen shows a Windows taskbar with the date and time '13:01 18/04/2026'.



Step :5 Access App

→ <http://3.87.125:30007>



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